Whole genome sequencing: what it adds A published Indian family: exome found nothing, genome found a deletion in BTK

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed article. Please read the original article (link below).

Source: Jain A, Govindaraj GM, Edavazhippurath A, Faisal N, Bhoyar RC, Gupta V, et al. *Whole genome sequencing identifies novel structural variant in a large Indian family affected with X-linked agammaglobulinemia.* PLoS ONE 2021;16(7):e0254407. doi:10.1371/journal.pone.0254407. CSIR-Institute of Genomics and Integrative Biology (New Delhi), Government Medical College Kozhikode, Apollo Hospitals Chennai, ICMR-NIIH Mumbai. Open access (CC BY 4.0). doi.org/10.1371/journal.pone.0254407

The family

- Extended Indian family; five affected boys
- Onset at 5-12 months: recurrent chest infections, diarrhoea, pyoderma, pyogenic meningitis
- Absent or atrophic tonsils, hypogammaglobulinaemia, very low or absent CD19 B cells
- All started on IVIG; bronchiectasis in two children without regular IVIG; one child died of meningitis at 20 months
- Mean age at diagnosis 44.6 months against onset at 7 months
- Pedigree: many boys who died early in life, attributed to infections

The genetic search

- Flow cytometry: reduced BTK expression on monocytes in one boy; mosaic pattern in his mother
- Exome in four boys: **no matching small variant or indel**
- Genome sequencing in two boys with structural-variant calling: **hemizygous 5,296 bp deletion removing BTK exons 3-5**
- On review, no reads over the region in the other boys' exome data
- Not found in 1000 Genomes, gnomAD or the Indian **IndiGen** databases
- Confirmed by **MLPA**

What it meant for the family

MLPA in **17 relatives** identified four affected males and **seven carrier women**, and showed several relatives were not carriers. Two boys underwent **haematopoietic stem cell transplant from matched sibling donors** and are reported disease-free. The authors explain that the exome missed the change possibly because of limited coverage and the **difficulty of calling structural variants** from exome data; genome sequencing identified the exact variant, enabling earlier diagnosis, genetic counselling, carrier identification and the option of prenatal diagnosis.

Talking points for the doctor visit

- A negative exome with a strong clinical and laboratory picture is not the end of the search.
- Genome sequencing detects small variants, copy number changes and structural variants in one test.
- Once the family variant is known, offer carrier testing to female relatives and counselling.
- Prenatal testing in India: written consent under the PCPNDT Act; disease testing only; fetal sex never disclosed.

EPISODE 9 · QUICK GUIDE

Whole genome sequencing: what it adds

Topic	Key points
What it reads	About 3 billion bases: coding and non-coding DNA. Exome reads only the coding parts (about 1-2%).
What it adds	No capture step, so coverage is more even, including over exons. Small variants and copy number changes in one test. Structural variants (inversions, translocation breakpoints), deep intronic and regulatory changes, mitochondrial DNA, and some repeat expansions with special analysis.
Evidence	Meta-analysis of 39 paediatric studies: pooled yield 38.6% for genome and 37.8% for exome; genome favoured on direct comparison (OR 1.54); usual care 7.8% (Nurchis et al. 2023).
When to consider	Exome negative with strong suspicion; suspected structural variant; one test instead of CMA plus exome; rapid genome for critically ill newborns in some centres. ACMG (2021) recommends exome or genome as a first- or second-tier test for congenital anomalies, developmental delay or intellectual disability.
Limits	Higher cost and very large data; non-coding changes are hard to interpret (more VUS); standard short-read genome can miss methylation disorders, low-level mosaicism and some difficult regions. Long-read sequencing is closing some gaps.
In practice	Trio gives a clearer report. Counsel first and take written consent on secondary findings. Data can be reanalysed later.
Sample	EDTA blood (parents too for a trio). Prenatal: amniotic fluid or CVS.

References

- Jain A et al. PLoS ONE 2021;16(7):e0254407. doi.org/10.1371/journal.pone.0254407
- Nurchis MC et al. Whole genome sequencing diagnostic yield for paediatric patients with suspected genetic disorders: systematic review, meta-analysis, and GRADE assessment. Arch Public Health 2023;81:93. doi:10.1186/s13690-023-01112-4
- Manickam K et al. Exome and genome sequencing for pediatric patients with congenital anomalies or intellectual disability: an evidence-based clinical guideline of the ACMG. Genet Med 2021;23:2029-2037.
- Pre-Conception and Pre-Natal Diagnostic Techniques (PCPNDT) Act, 1994 (India).

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